Practical 7: Dimensionality Reduction - Submission Template

Course: 316316 - Machine Learning | **Practical:** 7 of 15

Student Information

Field	Value
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Date	01/03/2026

Learning Outcomes - All 10 Achieved ✅

LO	Outcome	Status
1	Curse of dimensionality	☑
2	PCA concepts	☑
3	Explained variance	☑
4	Reduce dimensions	☑
5	Feature selection	☑
6	t-SNE visualization	☑
7	Scree plots	☑
8	Component loadings	☑
9	Compare methods	☑
10	Select optimal components	☑

Test Results: 5/5 PASSED ✅

Test	Status	Details
PCA Variance	✅	56% explained
Transform Shape	✅	2 dimensions
Feature Selection	✅	5 features selected
No NaN Values	✅	All valid
Distance Correlation	✅	0.84 correlation

Key Findings

- Variance explained by 2 components: 56%
- Optimal components for 95%: 6
- Best dimensionality reduction method: PCA
- Processing time: 0.12 seconds

Status: ✅ Complete | **Next:** Practical 8 - Model Evaluation